

Exploratory Data Analysis

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Project Title: Supply Chain Operations Analysis

Problem Statement

The rapid growth of the e-commerce industry has increased the complexity of supply chain and warehouse operations. Companies depend on efficient inventory management, reliable shipping operations, and product quality control to maintain customer satisfaction and reduce operational costs. However, operational inefficiencies such as delayed shipments, inventory imbalance, high transportation costs, and product defects continue to impact overall supply chain performance. This exploratory data analysis investigates supply chain and shipping datasets to identify relationships between operational variables and determine which factors most strongly affect business performance.

Business Impact

Exploring this data can help organizations improve supply chain efficiency by identifying operational bottlenecks and performance issues. The analysis can support better inventory management, reduce shipping delays, improve supplier quality, and optimize transportation costs. These improvements may lead to lower operational expenses, improved customer satisfaction, and stronger decision-making across warehouse and logistics operations.

General Dataset Information

In support of this analysis, the following datasets were used.

Dataset 1: Supply Chain Dataset

- **File Name:** Supply Chain Dataset.csv
- **Description:** Supply chain operational dataset containing information related to product sales, inventory management, shipping operations, supplier performance, transportation costs, stock levels, and defect rates.
- **Dataset Details:** Approximately 100+ Rows & 24 Columns
- **Size:** 21.05 KB
- **Source:** Kaggle - Supply Chain Dataset
<https://www.kaggle.com/datasets/amirmotefaker/supply-chain-dataset>

Dataset 2: E-Commerce Shipping Data

- **File Name:** E-Commerce Shipping Data.csv
- **Description:** Dataset containing information related to shipment performance, shipping modes, customer ratings, warehouse operations, and delivery outcomes.
- **Dataset Details:** Approximately 10,000+ Rows & 12 Columns

- **Size:** 440.46 KB
- **Source:** Kaggle – E-Commerce Shipping Data
<https://www.kaggle.com/datasets/prachi13/customer-analytics>

Target Features

The problem statement outlines several important metrics for evaluating supply chain performance and operational efficiency. The key features used for this analysis include:

- **Revenue generated:** Represents operational revenue generated from product sales.
- **Stock levels :** Measures inventory availability and warehouse efficiency.
- **Shipping times:** Indicates shipment and delivery performance.
- **Shipping costs:** Represents transportation and logistics expenses.
- **Defect rates:** Measures supplier and product quality performance.
- **Transportation modes:** Identifies operational differences between shipping methods.
- **Customer rating:** Measures customer satisfaction levels.
- **Reached on Time:** Indicates shipment delivery performance.
- **Product type:** Used to compare operational performance across product categories.

These features were used together to identify operational inefficiencies and evaluate relationships between shipping, inventory, supplier quality, and customer satisfaction.

List of Analysis

Analysis #1: Total Records and Dataset Overview

To provide context for the research effort, the datasets were examined to determine the number of operational records available for analysis. The Supply Chain dataset contains approximately 100+ operational records across 24 variables, while the E-Commerce Shipping dataset contains more than 10,000 shipment records across 12 variables. The large shipment dataset provides sufficient operational evidence to evaluate delivery performance, customer satisfaction, and warehouse efficiency.

Analysis #2: Shipping Time Distribution

Shipping times were analyzed to understand delivery performance across different transportation modes. The distribution revealed that most shipments were completed within moderate delivery ranges, while a smaller number of shipments experienced longer delays. Air transportation generally resulted in shorter shipping times compared to road and sea transportation.

The analysis suggests that transportation method has a significant impact on operational efficiency. Faster delivery methods improve customer satisfaction but may

also increase transportation costs. The findings support the hypothesis that shipping operations play an important role in overall supply chain performance.

Analysis #3: Revenue Distribution by Product Type

Revenue generated was grouped by product category to identify high-performing products within the supply chain system. Several product categories consistently generated higher revenue than others, while lower-performing categories contributed minimally to total sales.

The distribution was right-skewed, meaning a relatively small number of products generated a large portion of the total revenue. This suggests that organizations should focus on high-performing product categories to improve profitability and operational efficiency.

Analysis #4 - Inventory and Stock Level Analysis

Inventory availability and stock levels were examined to evaluate warehouse performance and inventory management practices. The analysis revealed large variation in stock levels across products. Some products maintained excessively high stock levels, while others showed low availability.

The imbalance in inventory distribution may indicate inefficient inventory planning. Overstocking increases holding costs, while low stock availability may result in delayed orders and lost sales opportunities. The findings support the importance of maintaining balanced inventory levels to improve operational efficiency.

Analysis #5 - Transportation Costs vs Revenue

Transportation costs were compared against revenue generated to determine whether higher logistics expenses improved business performance. The analysis revealed that higher transportation costs did not always result in higher revenue generation. This finding suggests that increasing shipping expenses alone does not guarantee improved operational outcomes. Companies must carefully balance transportation speed, delivery performance, and operational costs to maximize profitability.

Analysis #6 - Supplier Performance and Defect Rates

Supplier performance was analyzed using defect rates and inspection results. Most suppliers maintained relatively low defect percentages; however, several suppliers showed noticeably higher defect rates compared to others.

Higher defect rates negatively impact operational performance by increasing product returns, customer dissatisfaction, and operational waste. The analysis highlights the importance of supplier quality management within supply chain operations.

Analysis #7 - Customer Ratings and Delivery Performance

Customer ratings were compared against delivery outcomes to determine how shipping performance affects customer satisfaction. Shipments delivered on time generally received higher customer ratings than delayed shipments.

This relationship supports the hypothesis that delivery performance strongly influences customer satisfaction. Delayed shipments can reduce customer trust and increase customer service issues.

Analysis #8 - Basic Statistical Analysis

Basic statistical analysis was conducted on important operational variables such as revenue generated, stock levels, shipping times, shipping costs, and defect rates. The statistics revealed moderate to high variation across operational metrics.

Revenue generated showed significant variation between products, while shipping times and transportation costs demonstrated operational inconsistencies. Defect rates remained relatively low overall, although certain suppliers performed worse than others.

Conclusion

The exploratory data analysis identified several important operational relationships within supply chain and warehouse operations. Shipping efficiency, inventory management, supplier quality, and transportation costs all play significant roles in operational performance and customer satisfaction.

The findings support the original hypothesis that operational inefficiencies negatively impact supply chain performance. However, several analyses also revealed unexpected results. For example, higher transportation costs did not consistently improve revenue or customer satisfaction. This demonstrates that operational efficiency depends on balancing multiple factors rather than focusing on a single metric.

The EDA process also uncovered additional questions for future investigation, including whether certain warehouse locations contribute more heavily to shipping delays and whether specific product categories are more vulnerable to inventory shortages or defects. Overall, the analysis provides valuable insights that can support future dashboard development, KPI reporting, SQL analysis, and business recommendations.